

1. Split a list in two parts - sorted and unsorted
2. Pick the first element
3. Compare with all elements in the sorted sub-list
4. Shift all the elements in the sorted sub-list that is greater than the value to be sorted
5. Insert the value
6. Repeat until list is sorted

PROGRAM:

```
# Creating a function for Insertion
def Insertion_sort(lst):
```

```
# Outer loop to traverse through
  1 to len(lst)
```

```
for i in range(1, len(lst)):
```

```
    value = lst[i]
```

```
# Move elements of lst
```

```
# [0...i-1] that are
```

```
# greater than value to one
```

```
# position ahead of their
```

```
# current position
```

```
    j = i - 1
```

```
    while j >= 0 and value < lst[j]:
```

```
        lst[j+1] = lst[j]
```

```
        j -= 1
```

```
    lst[j+1] = value
```

```
    print("Intermediate lst is:",
          lst)
```

```
    return lst
```

```
# Driver code to test above
```

```
lst = []
```

```
n = int(input("Enter the number
of elements in the array"))
```

```
for i in range(n):
```

```
num = int(input("Enter number to be inserted in array"))
```

```
l1.append(num)
```

```
print("The unsorted list is:", l1)
```

```
print("The sorted list is:", insertion_sort(l1))
```

Output:

Enter the number of elements in the array : 3

Enter the number to be inserted in the array : 30

Enter the number to be inserted in the array : -30

Enter the number to be inserted in the array : 12

The unsorted list is : [30, -30, 12]

The Intermediate list is : [-30, 30, 12]

The Intermediate list is : [-30, 12, 30]

The sorted list is : [-30, 12, 30]

Result:

Program has been executed successfully.